CHAPTER – 16

C H A I N RULE

Differentiation Using the Chain Rule

1. Introduction to Chain Rule

Basic Idea

The Chain Rule is used to differentiate composite functions.
A composite function means one function inside another function.
Examples:

$$(x^2 + 1)^5$$

$$\sin(3x)$$

$$e^{x^2}$$

$$\sqrt{2x + 1}$$

2. Formula of Chain Rule

Chain Rule Formula

If

$$y = f(g(x))$$

then

$$\boxed{\frac{dy}{dx} = f'(g(x)) \cdot g'(x)}$$

Meaning

Derivative of outer function $\times$ Derivative of inner function

3. Steps to Apply Chain Rule

Procedure

1. Identify outer function
2. Identify inner function
3. Differentiate outer function
4. Keep inner function unchanged
5. Multiply by derivative of inner function

4. Basic Examples

Example 1

Differentiate:

$$y = (x^2 + 3)^4$$

Solution
Outer function:

$$u^4$$

Inner function:

$$u = x^2 + 3$$

Derivative:

$$\frac{dy}{dx} = 4(x^2 + 3)^3(2x)$$

$$\boxed{\frac{dy}{dx} = 8x(x^2 + 3)^3}$$

Example 2

Differentiate:

$$y = \sin(5x)$$

Solution
Outer function:

$$\sin u$$

Inner function:

$$u = 5x$$

Derivative:

$$\frac{dy}{dx} = \cos(5x) \cdot 5$$

$$\boxed{\frac{dy}{dx} = 5\cos(5x)}$$

Example 3

Differentiate:

$$y = e^{x^2}$$

Solution
Outer function:

$$e^u$$

Inner function:

$$u = x^2$$

Derivative:

$$\frac{dy}{dx} = e^{x^2}(2x)$$

$$\boxed{\frac{dy}{dx} = 2xe^{x^2}}$$

Example 4

Differentiate:

$$y = \sqrt{3x + 1}$$

Rewrite:

$$y = (3x + 1)^{1/2}$$

Derivative:

$$\frac{dy}{dx} = \frac{1}{2}(3x+1)^{-1/2}(3)$$

$$\boxed{\frac{dy}{dx} = \frac{3}{2\sqrt{3x+1}}}$$

5. Multiple Chain Rule

Nested Functions

Sometimes one function is inside another several times.
Example:

$$e^{(x^2+1)^3}$$

In such cases, apply chain rule repeatedly.

Example 5

Differentiate:

$$y = e^{(x^2+1)^3}$$

Solution
Outer function:

$$e^u$$

Middle function:

$$u = (x^2+1)^3$$

Inner function:

$$x^2+1$$

Derivative:

$$\frac{dy}{dx} = e^{(x^2+1)^3} \cdot 3(x^2+1)^2 \cdot 2x$$

$$\boxed{\frac{dy}{dx} = 6x(x^2+1)^2 e^{(x^2+1)^3}}$$

6. Important Formulas

Function	**Derivative**
$(u)^n$	$n(u)^{n-1} \cdot u'$
$\sin u$	$\cos u \cdot u'$
$\cos u$	$-\sin u \cdot u'$
e^u	$e^u \cdot u'$
$\ln u$	$\frac{u'}{u}$
$\sqrt{u}$	$\frac{u'}{2\sqrt{u}}$

7. Practice Questions

1. Differentiate:

$$y = (x^2 + 5)^6$$

2. Differentiate:

$$y = \cos(4x)$$

3. Differentiate:

$$y = e^{3x^2}$$

4. Differentiate:

$$y = \sqrt{x^2 + 1}$$

5. Differentiate:

$$y = \ln(2x + 1)$$

6. Differentiate:

$$y = \sin(x^3)$$

7. Differentiate:

$$y = (3x - 2)^7$$

8. Differentiate:

$$y = e^{\sin x}$$

9. Differentiate:

$$y = (\tan x)^5$$

10. Differentiate:

$$y = (x^2 + \sin x)^4$$

8. Answer Key

1. $$12x(x^2 + 5)^5$$

2. $$-4\sin(4x)$$

3. $$6xe^{3x^2}$$

4. $$\frac{x}{\sqrt{x^2+1}}$$

5. $$\frac{2}{2x+1}$$

6. $$3x^2\cos(x^3)$$

7. $$21(3x-2)^6$$

8. $$e^{\sin x}\cos x$$

9. $$5\tan^4 x \sec^2 x$$

10. $$4(x^2 + \sin x)^3(2x + \cos x)$$

Differentiation of Functions in Implicit Form

Implicit Differentiation

Sometimes y is not written directly as a function of x.
Instead, both variables appear together in one equation.
Examples:

$$x^2 + y^2 = 25$$

$$x^3 + y^3 = 6xy$$

Such differentiation is called:

$$\boxed{\text{Implicit Differentiation}}$$

1. Basic Idea of Implicit Differentiation

When differentiating with respect to x:

$$\frac{d}{dx}(y) = \frac{dy}{dx}$$

because y is treated as a function of x.

2. Steps for Implicit Differentiation

Procedure

1. Differentiate both sides with respect to x
2. Use chain rule whenever y appears
3. Collect all $\frac{dy}{dx}$ terms together
4. Solve for $\frac{dy}{dx}$

3. Basic Examples

Example 1

Differentiate:

$$x^2 + y^2 = 25$$

Solution
Differentiate both sides:

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$$

$$2x + 2y\frac{dy}{dx} = 0$$

Move terms:

$$2y\frac{dy}{dx} = -2x$$

Divide by $2y$:

$$\boxed{\frac{dy}{dx} = -\frac{x}{y}}$$

Example 2

Differentiate:

$$x^3 + y^3 = 6xy$$

Solution
Differentiate both sides:

$$3x^2 + 3y^2\frac{dy}{dx} = 6\left(x\frac{dy}{dx} + y\right)$$

Expand:

$$3x^2 + 3y^2\frac{dy}{dx} = 6x\frac{dy}{dx} + 6y$$

Collect $\frac{dy}{dx}$ terms:

$$3y^2\frac{dy}{dx} - 6x\frac{dy}{dx} = 6y - 3x^2$$

Factor:

$$\frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2$$

Thus,

$$\boxed{\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x}}$$

4. Implicit Differentiation with Trigonometric Functions

Example 3

Differentiate:

$$\sin(xy) = x + y$$

Solution
Differentiate both sides:

$$\cos(xy)\frac{d}{dx}(xy) = 1 + \frac{dy}{dx}$$

Using product rule:

$$\cos(xy)\left(x\frac{dy}{dx} + y\right) = 1 + \frac{dy}{dx}$$

Expand:

$$x\cos(xy)\frac{dy}{dx} + y\cos(xy) = 1 + \frac{dy}{dx}$$

Collect terms:

$$x\cos(xy)\frac{dy}{dx} - \frac{dy}{dx} = 1 - y\cos(xy)$$

Factor:

$$\frac{dy}{dx}\left(x\cos(xy) - 1\right) = 1 - y\cos(xy)$$

Therefore,

$$\boxed{\frac{dy}{dx} = \frac{1 - y\cos(xy)}{x\cos(xy) - 1}}$$

5. Implicit Differentiation with Exponential Functions

Example 4

Differentiate:

$$e^{xy} = x + y$$

Solution
Differentiate both sides:

$$e^{xy}\frac{d}{dx}(xy) = 1 + \frac{dy}{dx}$$

Using product rule:

$$e^{xy}\left(x\frac{dy}{dx}+y\right)=1+\frac{dy}{dx}$$

Expand:

$$xe^{xy}\frac{dy}{dx}+ye^{xy}=1+\frac{dy}{dx}$$

Collect terms:

$$xe^{xy}\frac{dy}{dx}-\frac{dy}{dx}=1-ye^{xy}$$

Factor:

$$\frac{dy}{dx}(xe^{xy}-1)=1-ye^{xy}$$

Thus,

$$\boxed{\frac{dy}{dx}=\frac{1-ye^{xy}}{xe^{xy}-1}}$$

6. Advanced Examples

Example 5

Differentiate:

$$x^2y+y^2=x$$

Solution
Differentiate both sides:

$$2xy+x^2\frac{dy}{dx}+2y\frac{dy}{dx}=1$$

Collect terms:

$$x^2\frac{dy}{dx}+2y\frac{dy}{dx}=1-2xy$$

Factor:

$$\frac{dy}{dx}(x^2+2y)=1-2xy$$

Therefore,

$$\boxed{\frac{dy}{dx}=\frac{1-2xy}{x^2+2y}}$$

Example 6

Differentiate:

$$\ln(xy) = x + y$$

Solution
Differentiate both sides:

$$\frac{1}{xy}\frac{d}{dx}(xy) = 1 + \frac{dy}{dx}$$

Using product rule:

$$\frac{x\frac{dy}{dx} + y}{xy} = 1 + \frac{dy}{dx}$$

Multiply by xy:

$$x\frac{dy}{dx} + y = xy + xy\frac{dy}{dx}$$

Collect terms:

$$x\frac{dy}{dx} - xy\frac{dy}{dx} = xy - y$$

Factor:

$$\frac{dy}{dx}(x - xy) = y(x - 1)$$

Thus,

$$\boxed{\frac{dy}{dx} = \frac{y(x-1)}{x(1-y)}}$$

7. Parametric Differentiation

Parametric Differentiation

Sometimes both x and y are expressed in terms of another variable t.
Example:

$$x = t^2, \qquad y = t^3$$

Such equations are called parametric equations.
The derivative is given by:

$$\boxed{\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}}$$

8. Steps for Parametric Differentiation

Procedure

1. Differentiate x with respect to t
2. Differentiate y with respect to t
3. Divide:

$$\frac{dy/dt}{dx/dt}$$

9. Basic Examples of Parametric Differentiation

Example 7

Given:

$$x = t^2, \qquad y = t^3$$

find

$$\frac{dy}{dx}$$

Solution
Differentiate:

$$\frac{dx}{dt} = 2t$$

$$\frac{dy}{dt} = 3t^2$$

Now,

$$\frac{dy}{dx} = \frac{3t^2}{2t}$$

$$\boxed{\frac{dy}{dx} = \frac{3t}{2}}$$

Example 8

Given:

$$x = \cos t, \qquad y = \sin t$$

find

$$\frac{dy}{dx}$$

Solution
Differentiate:

$$\frac{dx}{dt} = -\sin t$$

$$\frac{dy}{dt} = \cos t$$

Thus,

$$\frac{dy}{dx} = \frac{\cos t}{-\sin t}$$

$$\boxed{\frac{dy}{dx} = -\cot t}$$

10. Second Derivative in Parametric Form

Second Derivative Formula

If

$$\frac{dy}{dx} = f(t)$$

then

$$\boxed{\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}}$$

Example 9

Given:

$$x = t^2, \qquad y = t^3$$

find

$$\frac{d^2y}{dx^2}$$

Solution
First derivative:

$$\frac{dy}{dx} = \frac{3t}{2}$$

Differentiate with respect to t:

$$\frac{d}{dt}\left(\frac{3t}{2}\right) = \frac{3}{2}$$

Also,

$$\frac{dx}{dt} = 2t$$

Thus,

$$\frac{d^2y}{dx^2} = \frac{\frac{3}{2}}{2t}$$

$$\boxed{\frac{d^2y}{dx^2} = \frac{3}{4t}}$$

11. Advanced Parametric Example

Example 10

Given:

$$x = t - \sin t, \qquad y = 1 - \cos t$$

find

$$\frac{dy}{dx}$$

Solution
Differentiate:

$$\frac{dx}{dt} = 1 - \cos t$$

$$\frac{dy}{dt} = \sin t$$

Therefore,

$$\boxed{\frac{dy}{dx} = \frac{\sin t}{1 - \cos t}}$$

12. Important Formula Summary

Type	Formula
Implicit Differentiation	$\frac{d}{dx}(y) = \frac{dy}{dx}$
Parametric Differentiation	$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$
Second Derivative Parametric	$\frac{d^2y}{dx^2} = \frac{d/dt(dy/dx)}{dx/dt}$

13. Practice Questions

1. Differentiate implicitly:

$$x^2 + y^2 = 16$$

2. Differentiate:

$$x^3 + y^3 = 3xy$$

3. Differentiate:

$$\sin(x + y) = x$$

4. Differentiate:

$$e^{x+y} = xy$$

5. Given:

$$x = t^2, \qquad y = t^4$$

   find

$$\frac{dy}{dx}$$

6. Given:

$$x = \sin t, \qquad y = \cos t$$

   find

$$\frac{dy}{dx}$$

7. Find:

$$\frac{d^2y}{dx^2}$$

for:

$$x = t^3, \qquad y = t^2$$

8. Differentiate implicitly:

$$x \sin y = y \cos x$$

9. Differentiate:

$$\ln(x + y) = xy$$

10. Given:

$$x = e^t, \qquad y = e^{-t}$$

find

$$\frac{dy}{dx}$$

14. Answer Key

1. $$\frac{dy}{dx} = -\frac{x}{y}$$

2. $$\frac{dy}{dx} = \frac{y - x^2}{y^2 - x}$$

3. $$\frac{dy}{dx} = \sec(x + y) - 1$$

4. $$\frac{dy}{dx} = \frac{y - e^{x+y}}{e^{x+y} - x}$$

5. $$2t$$

6.
$$-\cot t$$

7.
$$-\frac{2}{9t^4}$$

8.
$$\frac{y\sin x+\sin y}{x\cos y-\cos x}$$

9.
$$\frac{xy-1/(x+y)}{1/(x+y)-x}$$

10.
$$-e^{-2t}$$

Logarithmic Differentiation

Introduction

Sometimes differentiation becomes difficult when:

- variables appear in both base and exponent,
- many functions are multiplied together,
- powers are complicated,
- roots and exponents occur simultaneously.

In such situations, we use:

$$\boxed{\text{Logarithmic Differentiation}}$$

1. Basic Idea of Logarithmic Differentiation

The method is based on taking logarithm on both sides of the equation.

If

$$y=f(x)$$

then take natural logarithm:

$$\ln y=\ln(f(x))$$

After simplifying, differentiate both sides implicitly.

2. Important Logarithmic Properties

Logarithmic Laws

$$\ln(ab) = \ln a + \ln b$$

$$\ln\left(\frac{a}{b}\right) = \ln a - \ln b$$

$$\ln(a^n) = n \ln a$$

3. Steps of Logarithmic Differentiation

Procedure

1. Assume:

$$y = f(x)$$

2. Take logarithm on both sides.
3. Use logarithmic laws to simplify.
4. Differentiate implicitly with respect to x.
5. Solve for:

$$\frac{dy}{dx}$$

4. Basic Examples

Example 1

Differentiate:

$$y = x^x$$

Solution

Take logarithm:

$$\ln y = \ln(x^x)$$

Using:

$$\ln(a^n) = n \ln a$$

$$\ln y = x \ln x$$

Differentiate both sides:

$$\frac{1}{y}\frac{dy}{dx} = x\left(\frac{1}{x}\right) + \ln x$$

$$\frac{1}{y}\frac{dy}{dx} = 1 + \ln x$$

Multiply by y:

$$\frac{dy}{dx} = y(1 + \ln x)$$

Substitute $y = x^x$:

$$\boxed{\frac{dy}{dx} = x^x(1 + \ln x)}$$

Example 2

Differentiate:

$$y = (x^2 + 1)^x$$

Solution
Take logarithm:

$$\ln y = x \ln(x^2 + 1)$$

Differentiate:

$$\frac{1}{y}\frac{dy}{dx} = \ln(x^2 + 1) + x\left(\frac{2x}{x^2 + 1}\right)$$

Thus,

$$\frac{dy}{dx} = y\left[\ln(x^2 + 1) + \frac{2x^2}{x^2 + 1}\right]$$

Substitute $y = (x^2 + 1)^x$:

$$\boxed{\frac{dy}{dx} = (x^2 + 1)^x\left[\ln(x^2 + 1) + \frac{2x^2}{x^2 + 1}\right]}$$

5. Logarithmic Differentiation for Products

Example 3

Differentiate:

$$y = x^2(x+1)^3(x^2+1)^4$$

Solution
Take logarithm:

$$\ln y = \ln(x^2) + \ln((x+1)^3) + \ln((x^2+1)^4)$$

Using logarithmic laws:

$$\ln y = 2\ln x + 3\ln(x+1) + 4\ln(x^2+1)$$

Differentiate:

$$\frac{1}{y}\frac{dy}{dx} = \frac{2}{x} + \frac{3}{x+1} + \frac{8x}{x^2+1}$$

Multiply by y:

$$\boxed{\frac{dy}{dx} = x^2(x+1)^3(x^2+1)^4\left(\frac{2}{x} + \frac{3}{x+1} + \frac{8x}{x^2+1}\right)}$$

6. Logarithmic Differentiation for Quotients

Example 4

Differentiate:

$$y = \frac{(x+1)^3\sqrt{x^2+1}}{x^4}$$

Solution
Take logarithm:

$$\ln y = 3\ln(x+1) + \frac{1}{2}\ln(x^2+1) - 4\ln x$$

Differentiate:

$$\frac{1}{y}\frac{dy}{dx} = \frac{3}{x+1} + \frac{1}{2}\cdot\frac{2x}{x^2+1} - \frac{4}{x}$$

$$= \frac{3}{x+1} + \frac{x}{x^2+1} - \frac{4}{x}$$

Thus,

$$\boxed{\frac{dy}{dx} = \frac{(x+1)^3\sqrt{x^2+1}}{x^4}\left(\frac{3}{x+1} + \frac{x}{x^2+1} - \frac{4}{x}\right)}$$

7. Logarithmic Differentiation with Trigonometric Functions

Example 5

Differentiate:

$$y = (\sin x)^x$$

Solution
Take logarithm:

$$\ln y = x\ln(\sin x)$$

Differentiate:

$$\frac{1}{y}\frac{dy}{dx} = \ln(\sin x) + x\left(\frac{\cos x}{\sin x}\right)$$

$$= \ln(\sin x) + x\cot x$$

Thus,

$$\boxed{\frac{dy}{dx} = (\sin x)^x\left(\ln(\sin x) + x\cot x\right)}$$

8. Logarithmic Differentiation with Exponential Functions

Example 6

Differentiate:

$$y = (e^x)^{\sin x}$$

Solution
Take logarithm:

$$\ln y = \sin x \cdot \ln(e^x)$$

Since

$$\ln(e^x) = x$$

$$\ln y = x \sin x$$

Differentiate:

$$\frac{1}{y}\frac{dy}{dx} = x\cos x + \sin x$$

Thus,

$$\boxed{\frac{dy}{dx} = (e^x)^{\sin x}(x\cos x + \sin x)}$$

9. Advanced Examples

Example 7

Differentiate:

$$y = \left(\frac{x^2+1}{x-1}\right)^x$$

Solution
Take logarithm:

$$\ln y = x \ln\left(\frac{x^2+1}{x-1}\right)$$

Using logarithmic law:

$$\ln y = x[\ln(x^2+1) - \ln(x-1)]$$

Differentiate:

$$\frac{1}{y}\frac{dy}{dx} = \ln\left(\frac{x^2+1}{x-1}\right) + x\left(\frac{2x}{x^2+1} - \frac{1}{x-1}\right)$$

Thus,

$$\boxed{\frac{dy}{dx} = \left(\frac{x^2+1}{x-1}\right)^x \left[\ln\left(\frac{x^2+1}{x-1}\right) + x\left(\frac{2x}{x^2+1} - \frac{1}{x-1}\right)\right]}$$

Example 8

Differentiate:

$$y = (x + \sqrt{x^2+1})^x$$

Solution
Take logarithm:

$$\ln y = x \ln(x + \sqrt{x^2+1})$$

Differentiate:

$$\frac{1}{y}\frac{dy}{dx} = \ln(x + \sqrt{x^2+1}) + x\left[\frac{1 + \frac{x}{\sqrt{x^2+1}}}{x + \sqrt{x^2+1}}\right]$$

Thus,

$$\boxed{\frac{dy}{dx} = (x + \sqrt{x^2+1})^x \left[\ln(x + \sqrt{x^2+1}) + x\left(\frac{1 + \frac{x}{\sqrt{x^2+1}}}{x + \sqrt{x^2+1}}\right)\right]}$$

10. Important Observations

When to Use Logarithmic Differentiation

Use logarithmic differentiation when:

- variable appears in exponent,

$$x^x, \quad (\sin x)^x$$

- many complicated products occur,
- roots and powers appear together,
- expressions become difficult using ordinary differentiation.

11. Important Formula Summary

Function	**Derivative**
x^x	$x^x(1+\ln x)$
$(u(x))^{v(x)}$	$u^v\left[v'\ln u + v\frac{u'}{u}\right]$
$\ln y$	$\frac{1}{y}\frac{dy}{dx}$

12. Practice Questions

1. Differentiate:

$$y = x^x$$

2. Differentiate:

$$y = (x^2+1)^x$$

3. Differentiate:

$$y = (\cos x)^x$$

4. Differentiate:

$$y = x^2(x+1)^3$$

5. Differentiate:

$$y = \frac{(x+1)^2}{x^3}$$

6. Differentiate:

$$y = (e^x)^x$$

7. Differentiate:

$$y = (\tan x)^x$$

8. Differentiate:

$$y = \left(\frac{x+1}{x-1}\right)^x$$

9. Differentiate:

$$y = (x^2 + \sin x)^x$$

10. Differentiate:

$$y = (\ln x)^x$$

13. Answer Key

1.
$$x^x(1 + \ln x)$$

2.
$$(x^2+1)^x \left[\ln(x^2+1) + \frac{2x^2}{x^2+1}\right]$$

3.
$$(\cos x)^x \left[\ln(\cos x) - x\tan x\right]$$

4.
$$x^2(x+1)^3 \left(\frac{2}{x} + \frac{3}{x+1}\right)$$

5.
$$\frac{(x+1)^2}{x^3}\left(\frac{2}{x+1} - \frac{3}{x}\right)$$

6.
$$(e^x)^x(x+1)$$

7.
$$(\tan x)^x \left[\ln(\tan x) + x\frac{\sec^2 x}{\tan x}\right]$$

8.
$$\left(\frac{x+1}{x-1}\right)^x \left[\ln\left(\frac{x+1}{x-1}\right) + x\left(\frac{1}{x+1} - \frac{1}{x-1}\right)\right]$$

9.
$$(x^2+\sin x)^x \left[\ln(x^2+\sin x) + x\frac{2x+\cos x}{x^2+\sin x}\right]$$

10.
$$(\ln x)^x \left[\ln(\ln x) + \frac{1}{\ln x}\right]$$

Successive Differentiation

Introduction

The derivative of a function can itself be differentiated again.
This process is called:

$$\boxed{\text{Successive Differentiation}}$$

The first derivative is written as:

$$\frac{dy}{dx}$$

The second derivative is:

$$\frac{d^2y}{dx^2}$$

The third derivative is:

$$\frac{d^3y}{dx^3}$$

and so on.

1. Notations

Derivative	**Notation**
First Derivative	$\frac{dy}{dx},\ y'$
Second Derivative	$\frac{d^2y}{dx^2},\ y''$
Third Derivative	$\frac{d^3y}{dx^3}$
n^{th} Derivative	$\frac{d^ny}{dx^n}$

2. Examples of Successive Differentiation

Example 1

Find first and second derivatives of:

$$y = x^3 - 2x^2 + 5x + 1$$

Solution

First derivative:

$$\frac{dy}{dx} = 3x^2 - 4x + 5$$

Second derivative:

$$\frac{d^2y}{dx^2} = 6x - 4$$

$$\boxed{\frac{dy}{dx} = 3x^2 - 4x + 5}$$

$$\boxed{\frac{d^2y}{dx^2} = 6x - 4}$$

Example 2

Find the third derivative of:

$$y = x^4$$

Solution
First derivative:

$$y' = 4x^3$$

Second derivative:

$$y'' = 12x^2$$

Third derivative:

$$y''' = 24x$$

$$\boxed{\frac{d^3y}{dx^3} = 24x}$$

Example 3

Find the second derivative of:

$$y = \sin x$$

Solution
First derivative:

$$y' = \cos x$$

Second derivative:

$$y'' = -\sin x$$

$$\boxed{\frac{d^2y}{dx^2} = -\sin x}$$

3. Geometrical Interpretation of Derivative

Geometrical Meaning

The derivative of a function at a point gives:

$$\boxed{\text{Slope of the tangent to the curve}}$$

If

$$y = f(x)$$

then

$$\frac{dy}{dx}$$

represents the slope of the tangent at that point.

4. Slope of Tangent

For a curve:

$$y = f(x)$$

the slope at point (x, y) is:

$$m = \frac{dy}{dx}$$

Example 4

Find slope of tangent to:

$$y = x^2$$

at $x = 2$.

Solution

Differentiate:

$$\frac{dy}{dx} = 2x$$

At $x = 2$:

$$m = 2(2) = 4$$

$$\boxed{\text{Slope} = 4}$$

5. Tangent and Normal

Tangent and Normal

Tangent:
A line touching the curve at a point is called tangent.

Normal:
A line perpendicular to tangent is called normal.

6. Equations of Tangent and Normal

For point:

$$(x_1, y_1)$$

Slope of tangent:

$$m = \left(\frac{dy}{dx}\right)_{(x_1,y_1)}$$

Equation of tangent:

$$\boxed{y - y_1 = m(x - x_1)}$$

Slope of normal:

$$-\frac{1}{m}$$

Equation of normal:

$$\boxed{y - y_1 = -\frac{1}{m}(x - x_1)}$$

Example 5

Find tangent and normal to:

$$y = x^2$$

at point $(1, 1)$.
Solution
Differentiate:

$$\frac{dy}{dx} = 2x$$

At $x = 1$:

$$m = 2$$

Equation of tangent:

$$y - 1 = 2(x - 1)$$

$$\boxed{y = 2x - 1}$$

Slope of normal:

$$-\frac{1}{2}$$

Equation of normal:

$$y - 1 = -\frac{1}{2}(x - 1)$$

$$\boxed{y = -\frac{1}{2}x + \frac{3}{2}}$$

7. Maxima and Minima

Definition

A function has:

- **Maximum value** if function changes from increasing to decreasing.
- **Minimum value** if function changes from decreasing to increasing.

These points are called:

$\boxed{\text{Turning Points or Critical Points}}$

8. Critical Points

Critical points occur where:

$$\boxed{\frac{dy}{dx} = 0}$$

or derivative does not exist.

9. Second Derivative Test

Second Derivative Test

Suppose:

$$\frac{dy}{dx} = 0$$

at $x = a$.
Then:

- If

$$\frac{d^2y}{dx^2} > 0$$

  then function has minimum value.

- If

$$\frac{d^2y}{dx^2} < 0$$

  then function has maximum value.

- If

$$\frac{d^2y}{dx^2} = 0$$

  test fails.

10. Examples of Maxima and Minima

Example 6

Find maxima or minima of:

$$y = x^2 - 4x + 1$$

Solution
First derivative:

$$\frac{dy}{dx} = 2x - 4$$

Set equal to zero:

$$2x - 4 = 0$$

$$x = 2$$

Second derivative:

$$\frac{d^2y}{dx^2} = 2$$

Since:

$$2 > 0$$

function has minimum value at $x = 2$.
Find minimum value:

$$y = (2)^2 - 4(2) + 1$$

$$= 4 - 8 + 1 = -3$$

$$\boxed{\text{Minimum value} = -3}$$

Example 7

Find maxima or minima of:

$$y = -x^2 + 6x - 5$$

Solution
First derivative:

$$\frac{dy}{dx} = -2x + 6$$

Set equal to zero:

$$-2x + 6 = 0$$

$$x = 3$$

Second derivative:

$$\frac{d^2y}{dx^2} = -2$$

Since:

$$-2 < 0$$

function has maximum value.
Maximum value:

$$y = -(3)^2 + 6(3) - 5$$

$$= -9 + 18 - 5$$

$$= 4$$

$$\boxed{\text{Maximum value} = 4}$$

11. Advanced Examples

Example 8

Find tangent to:

$$y = \sin x$$

at

$$x = \frac{\pi}{4}$$

Solution
Differentiate:

$$\frac{dy}{dx} = \cos x$$

At

$$x = \frac{\pi}{4}$$

$$m = \frac{\sqrt{2}}{2}$$

Point on curve:

$$\left(\frac{\pi}{4}, \frac{\sqrt{2}}{2}\right)$$

Equation:

$$y - \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2}\left(x - \frac{\pi}{4}\right)$$

Example 9

Find second derivative of:

$$y = e^x$$

Solution
First derivative:

$$y' = e^x$$

Second derivative:

$$y'' = e^x$$

$$\boxed{\frac{d^2y}{dx^2} = e^x}$$

Example 10

Find critical points of:

$$y = x^3 - 3x^2 + 2$$

Solution
Differentiate:

$$\frac{dy}{dx} = 3x^2 - 6x$$

Set equal to zero:

$$3x(x - 2) = 0$$

$$x = 0, \quad x = 2$$

Second derivative:

$$\frac{d^2y}{dx^2} = 6x - 6$$

At $x = 0$:

$$-6 < 0$$

Maximum point.

At $x = 2$:

$$6 > 0$$

Minimum point.

$$\boxed{x = 0 \text{ maximum}, \quad x = 2 \text{ minimum}}$$

12. Summary Table

Concept	**Formula**
First Derivative	$\frac{dy}{dx}$
Second Derivative	$\frac{d^2y}{dx^2}$
Slope of Tangent	$\frac{dy}{dx}$
Equation of Tangent	$y - y_1 = m(x - x_1)$
Equation of Normal	$y - y_1 = -\frac{1}{m}(x - x_1)$
Critical Point	$\frac{dy}{dx} = 0$
Minimum Condition	$\frac{d^2y}{dx^2} > 0$
Maximum Condition	$\frac{d^2y}{dx^2} < 0$

13. Practice Questions

1. Find first and second derivatives of:

$$y = x^4 - 2x^2 + 1$$

2. Find third derivative of:

$$y = x^5$$

3. Find slope of tangent to:

$$y = x^2 + 1$$

at $x = 3$.

4. Find tangent and normal to:

$$y = x^2$$

at point $(2, 4)$.

5. Find maxima or minima of:

$$y = x^2 - 6x + 5$$

6. Find maxima or minima of:

$$y = -x^2 + 8x - 3$$

7. Find second derivative of:

$$y = \cos x$$

8. Find tangent to:

$$y = e^x$$

at $x = 0$.

9. Find critical points of:

$$y = x^3 - 6x^2 + 9x$$

10. Find normal to:

$$y = x^3$$

at point $(1, 1)$.

14. Answer Key

1.
$$y' = 4x^3 - 4x$$
$$y'' = 12x^2 - 4$$

2.
$$60x^2$$

3.
$$6$$

4. Tangent:
$$y = 4x - 4$$
Normal:
$$y = -\frac{1}{4}x + \frac{9}{2}$$

5. Minimum value:
$$-4$$

6. Maximum value:
$$13$$

7.
$$-\cos x$$

8.
$$y = x + 1$$

9.
$$x = 1, \quad x = 3$$

10.
$$y = -\frac{1}{3}x + \frac{4}{3}$$